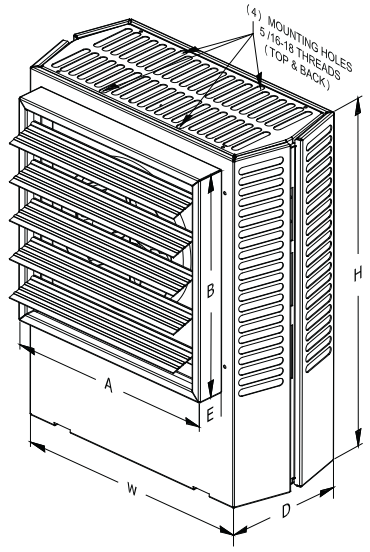
# Markel Products Company

MADE IN THE U.S.A.

## 5100 SERIES
HORIZONTAL / VERTICAL UNIT HEATER

MARKEL

| | | | |
|---|---|---|---|
| **CUSTOMER** | | **DATE:** | |
| **PROJECT** | | | |
| **LOCATION** | | | |
| **ARCHITECT** | | | |
| **ENGINEER** | | | |
| **CONTRACTOR** | | | |
| **SUBMITTED BY** | | | |
| **APPROVED BY** | | | |
| **APPROVED BY** | | | |




## SUBMITTAL DATA SHEET

| QTY | MODEL | TAG | WATTAGE | VOLTS | PHASE | AMPS | DISCONNECT | THERMOSTAT | BRACKET | OPTIONS |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | |
| | | | | | | | | | | |
| | | | | | | | | | | |
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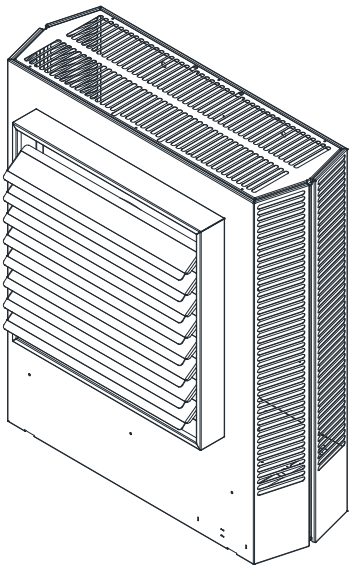
**NOTES/SPECIAL INSTRUCTIONS:**

PRINT FORM  EMAIL FORM  CLEAR FORM

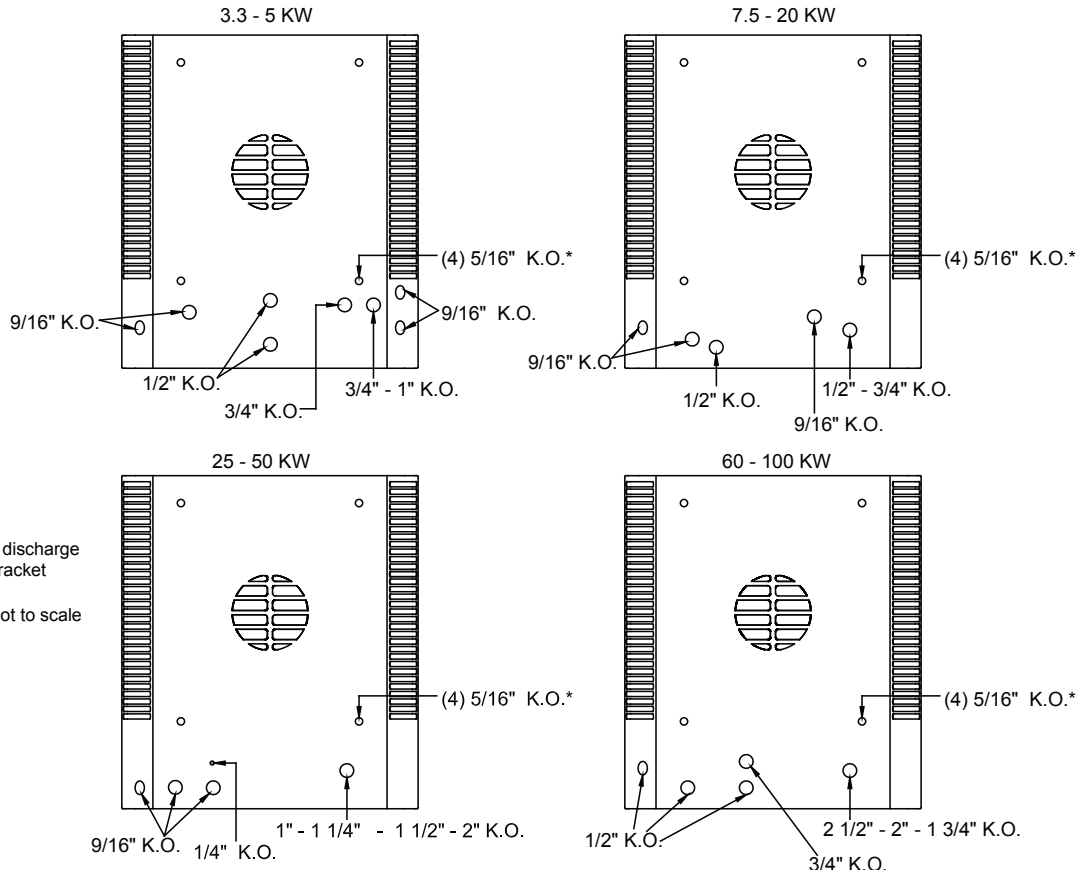
## *Product Dimensions*

| UNIT SIZE (KW) | DIMENSIONS (INCHES) | | | | | | WEIGHT (LBS) |
|---|---|---|---|---|---|---|---|
| | H | W | D | E | A | B | |
| 3.3 & 5 | 17 ¾ | 14 15/32 | 6 ½ | 1 5/8 | 11 ½ | 11 ½ | 25 |
| 7.5 & 10 | 24 5/16 | 21 ½ | 6 ½ | 2 7/32 | 15 ¼ | 15 ¼ | 54 |
| 15 & 20 | 28 11/16 | 21 ½ | 6 ½ | 2 7/32 | 15 ¼ | 15 ¼ | 65 |
| 25 THRU 50 | 34 | 29 ¼ | 10 1/16 | 2 5/8 | 20 ¼ | 20 ¼ | 120 |
| 60 & 70 | 35 ½ | 30 ½ | 14 ½ | 5 | 23 ½ | 23 ½ | 141 |
| 80 - 90 - 100 | 39 ½ | 37 5/8 | 18 ½ | 5 | 27 ½ | 27 ½ | 202 |

# *Product Specifications*



- Heavy 18-gauge steel construction
- Powder coated paint finish
- Copper clad steel sheath element with brazed fins and
- Nichrome element wire
- Totally enclosed fan motors
- Louvers are individually adjustable for directional air flow
- Optional remote and unit mounted controls include thermostats, disconnects, and fan switches
- Optional accessories include mounting brackets, louvers, fan guards, and dust shields




* For vertical discharge mounting bracket

Diagrams not to scale

HOW TO DETERMINE A MODEL:

| HF | 2 | B | 51 | 10 | C | A | 1 |
|---|---|---|---|---|---|---|---|

| Element Volts | Phase | Motor Voltage | Model Series | Element KW | Control System | Transformer | Control Volts |
|---|---|---|---|---|---|---|---|
| F = 208 | 1 = 1-Phase | F = 208 | 51 | | Blank = None | Blank = None | 1 = 24 |
| H = 240 | 2 = 1 or 3-Ph. | H = 240 | | | C = Contactor | A = Included | 2 = 120 |
| HF = 240/208 | 3 = 3-Phase | B = 240 / 208 | | | | | (with CA option) |
| G = 277 | | G = 277 | | | | | |
| P = 480 | | P = 480 | | | | | |

* 600 Volt units 5KW - 50KW available on special order, consult factory for pricing & availability

# *Product Specifications*

**CONSTRUCTION:**
Heavy 18 gauge welded steel cabinet with powder coated finish and control compartment housing a master terminal board with a hinged and latched access door, simplifying wiring, installation & maintenance.

**HEATING ELEMENT:**
Copper clad steel sheath element with continuously brazed steel fins formed to allow side draw through air flow.

**OVERHEAT PROTECTION:**
All units come equipped with automatic reset type limit controls to de-energize the heater should an over-temperature situation occur.

**FAN and MOTOR:**
Totally enclosed, 1-speed, 1-phase, permanently lubricated, thermally protected motors with unit bearings on 3 KW - 20 KW models. Totally enclosed, 2-speed, 1-phase, permanently lubricated, thermally protected motors with sleeve bearings on 25 KW - 50 KW models. All motors mounted with rubber insulators to minimize vibration & noise. Fan over-ride purges unit of residual heat at shutdown.

**LOUVER ASSEMBLY:**
Louvers are individually adjustable for directional control of air flow up to 15° from straight horizontal. Optional diffusers available for down flow (vertical discharge) applications.

**TEMPERATURE CONTROLS:**
Optional low voltage and line voltage thermostats available with an adjustable temperature range of 40°F to 110°F. Units with model numbers ending in CA1 are factory wired for low voltage controls. 25 KW through 50 KW units are designed for two stage heating operation.

**INSTALLATION:**
Unit Heaters can be mounted for horizontal or vertical discharge. Applications up to 6000 Ft. See UH Series above 6000 Ft.

**FEATURES:**
- Made in U.S.A.
- 3.3 kW - 50 kW models are cULus listed
- 55 kW - 100 kW models are cETLus listed

**FIELD INSTALLED OPTIONS:**
- In-unit or wall mounted temperature control thermostats low or line voltage.
- Summer fan switch to operate the fan only.
- Power disconnect switch.
- Heat stratification thermostat.

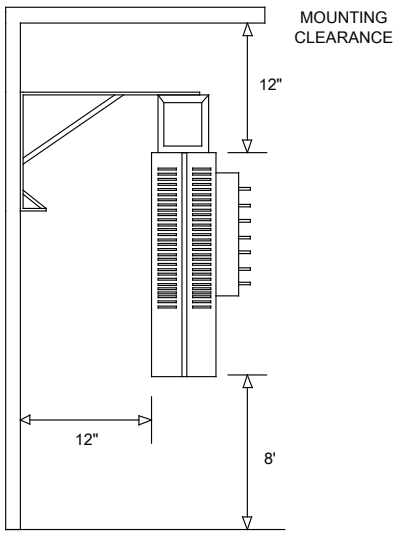


FIG. 1
HORIZONTAL DISCHARGE

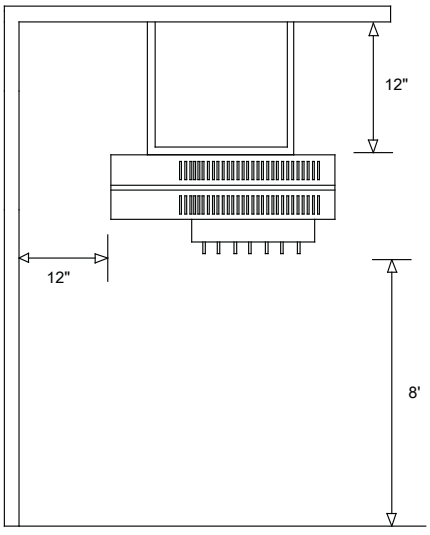


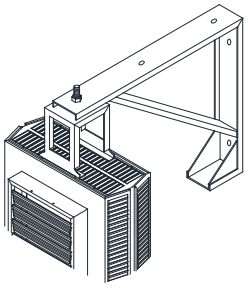
FIG. 2
VERTICAL DISCHARGE

# *Standard Models 3kW - 100kW*

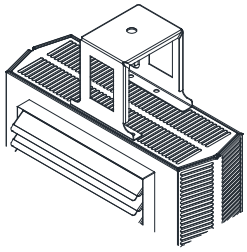
| MFG CATALOG NUMBER | MFG MODEL NUMBER | KW | BTU / H | VOLTS | PH | AMPS | CONTROL VOLTAGE | TEMP RISE | AIR THROW | CFM | RECOMMENDED MOUNTING HT. | | WT. (LBS) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Horizontal | Vertical | |
| 06450802 | F1F5103N | 3.3 | 11.2 | 208 | 1 | 15.9 / 9.17 | 208 | 26°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06451002 | HF1B5103N | 3.3/2.5 | 11.2 / 8.5 | 240/208 | 1 | 13.7 / 11.9 | 240 / 208 | 26°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06456802 | F2F5103N | 3.3 | 11.2 | 208 | 1 / 3 | 15.9 / 9.17 | 208 | 26°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06457002 | HF2B5103N | 3.3/2.5 | 11.2 / 8.5 | 240/208 | 1 / 3 | 13.7 / 11.9 | 240 / 208 | 26°F | 12′ | 400 | 9′ | 9′ | 25 |
| | HF2B5103N | 3.3/2.5 | 11.2 / 8.5 | 240/208 | 3 | 7.9 / 6.9 | 240 / 208 | 26°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06457202 | G1G5103N | 3.3 | 11.2 | 277 | 1 | 11.9 | 277 | 26°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06451202 | P3P5103CA1N* | 3.3 | 11.2 | 480 | 3 | 4.0 | 24 | 26°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06455402 | F1F5105N | 5.0 | 17.1 | 208 | 1 | 24.1 | 208 | 40°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06455602 | HF1B5105N | 5.0/3.7 | 17.1 / 12.8 | 240/208 | 1 | 20.9 / 18.1 | 240 / 208 | 40°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06451402 | F2F5105N | 5.0 | 17.1 | 208 | 1 / 3 | 24.1 | 208 | 40°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06451402 | F2F5105N | 5.0 | 17.1 | 208 | 3 | 13.9 | 208 | 40°F | 12′ | 400 | 9′ | 9′ | 25 |
| 06451602 | HF2B5105N | 5.0 | 17.1 | 240 | 1 / 3 | 20.8 / 18.1 | 240 | 40°F | 12′ | 400 | 9′ | 9′ | 27 |
| 06451602 | HF2B5105N | 3.7 | 12.8 | 208 | 1 / 3 | 17.1 / 10.4 | 208 | 40°F | 12′ | 400 | 9′ | 9′ | 27 |
| 06458402 | G1G5105N | 5.0 | 17.1 | 277 | 1 | 18.1 | 277 | 40°F | 12′ | 400 | 9′ | 9′ | 27 |
| 06451802 | P3P5105CA1N* | 5.0 | 17.1 | 480 | 3 | 6.1 | 24 | 40°F | 12′ | 400 | 9′ | 9′ | 27 |
| 06452002 | F2F5107CA1L* | 7.5 | 25.6 | 208 | 1 / 3 | 36.1 / 20.8 | 24 | 34°F | 22′ | 700 | 10′ | 12′ | 54 |
| 06452202 | HF2B5107CA1L* | 7.5 | 25.6 | 240 | 1 / 3 | 27.1 / 31.1 | 24 | 34°F | 22′ | 700 | 10′ | 12′ | 54 |
| 06452202 | HF2B5107CA1L* | 5.6 | 19.2 | 208 | 1 / 3 | 31.3 / 27.1 | 24 | 34°F | 22′ | 700 | 10′ | 12′ | 54 |
| 06459202 | G1G5107CA1L* | 7.5 | 25.6 | 277 | 1 | 27.1 | 24 | 34°F | 22′ | 700 | 10′ | 12′ | 54 |
| 06452402 | P3P5107CA1N* | 7.5 | 25.6 | 480 | 3 | 9.1 | 24 | 34°F | 22′ | 700 | 10′ | 12′ | 54 |
| 06452602 | F2F5110CA1L* | 9.9 | 33.8 | 208 | 1 / 3 | 47.8 / 27.4 | 24 | 45°F | 22′ | 700 | 10′ | 14′ | 55 |
| 06452802 | HF2B5110CA1L* | 10.0 | 34.1 | 240 | 1 / 3 | 41.2 / 24.0 | 24 | 45°F | 22′ | 700 | 10′ | 14′ | 55 |
| 06452802 | HF2B5110CA1L* | 7.5 | 25.6 | 208 | 1 / 3 | 36.1 / 20.8 | 24 | 45°F | 22′ | 700 | 10′ | 14′ | 55 |
| 06456402 | G1G5110CA1L* | 10.0 | 34.1 | 277 | 1 | 36.1 | 24 | 45°F | 22′ | 700 | 10′ | 14′ | 55 |
| 06453002 | P3P5110CA1N* | 10.0 | 34.1 | 480 | 3 | 12.4 | 24 | 45°F | 22′ | 700 | 10′ | 14′ | 55 |
| 06453202 | F3F5115CA1L* | 15.0 | 51.2 | 208 | 3 | 41.7 | 24 | 45°F | 22′ | 1100 | 10′ | 14′ | 64 |
| 06453402 | HF3B5115CA1L* | 15.0/11.2 | 51.2 / 38.4 | 240/208 | 3 | 36.1 / 31.3 | 24 | 43°F | 32′ | 1100 | 11′ | 20′ | 64 |
| 06453602 | P3P5115CA1N* | 15.0 | 51.2 | 480 | 3 | 18.1 | 24 | 43°F | 32′ | 1100 | 11′ | 20′ | 64 |
| 06453802 | HF3B5120CA1L* | 19.7/14.8 | 67.2 / 50.5 | 240/208 | 3 | 47.8 / 41.1 | 24 | 57°F | 32′ | 1100 | 12′ | 18′ | 65 |
| 06454002 | P3P5120CA1N* | 20.0 | 68.3 | 480 | 3 | 24.1 | 24 | 57°F | 32′ | 1100 | 12′ | 18′ | 65 |
| 06458802 | F3F5125CA1L* | 25.0 | 85.3 | 208 | 3 | 69.5 | 24 | 57°F | 32′ | 2000/1800 | 12′ | 18′ | 65 |
| 06459802 | P3P5125CA1N* | 25.0 | 85.3 | 480 | 3 | 30.1 | 24 | 57°F | 32′ | 2000/1801 | 12′ | 18′ | 65 |
| 06454202 | F3F5130CA1L* | 30.0 | 102.4 | 208 | 3 | 83.4 | 24 | 57°F | 32′ | 2000/1802 | 12′ | 18′ | 65 |
| 06454402 | HF3B5130CA1L* | 30.0/22.5 | 102.4 / 76.8 | 240/208 | 3 | 72.3 / 62.5 | 24 | 47/53°F | 40′ | 2000/1803 | 12′ | 20′ | 120 |
| 06454602 | P3P5130CA1N* | 30.0 | 102.4 | 480 | 3 | 36.2 | 24 | 47/53°F | 40′ | 2000/1804 | 12′ | 20′ | 120 |
| 06440402 | F3F5140CA1L* | 40.0 | 136.5 | 208 | 3 | 111.2 | 24 | 47/53°F | 40′ | 2000/1805 | 12′ | 20′ | 120 |
| 06440602 | HF3B5140CA1L* | 40.0/30.0 | 136.5/102.4 | 240/208 | 3 | 96.4 / 83.4 | 24 | 40/45°F | 55′ | 3100/2800 | 15′ | 24′ | 120 |
| 06440802 | P3P5140CA1N* | 39.0 | 133.1 | 480 | 3 | 47.0 | 24 | 40/45°F | 55′ | 3100/2801 | 15′ | 24′ | 120 |
| 06454802 | F3F5150CA1L* | 49.6 | 169.3 | 208 | 3 | 139.0 | 24 | 40/45°F | 55′ | 3100/2802 | 15′ | 24′ | 120 |
| 06455002 | HF3B5150CA1L* | 50.0/37.5 | 170.6/128.0 | 240/208 | 3 | 120.5/104.3 | 24 | 51/56°F | 50′ | 3100/2800 | 15′ | 22′ | 120 |
| 06455202 | P3P5150CA1N | 50.0 | 170.6 | 480 | 3 | 60.3 | 24 | 51/56°F | 50′ | 3100/2801 | 15′ | 22′ | 120 |

| MFG CATALOG NUMBER | MFG MODEL NUMBER | KW | BTUs / HR (x 1000) | VOLTS | PHASE | CONTROL VOLTAGE | AMPS / PHASE | BRANCH CIRCUIT PROTECTION SIZE |
|---|---|---|---|---|---|---|---|---|
| 07213402 | F3F5160CA1 | 60 | 204.7 | 208 | 3 | 24 | 166.7 | 225 |
| 07213502 | HF3B5160CA1 | 60/45 | 204.7/153.6 | 240/208 | 3 | 24 | 144.6/125 | 200/175 |
| 07213602 | P3P5160CA1 | 60 | 204.7 | 480 | 3 | 24 | 72.3 | 100 |
| 07214202 | F3F5170CA1 | 70 | 238.9 | 208 | 3 | 24 | 194.4 | 250 |
| 07214302 | HF3B5170CA1 | 70/52.5 | 238.9/179.2 | 240/208 | 3 | 24 | 168.7/145.8 | 225/200 |
| 07214402 | P3P5170CA1 | 70 | 238.9 | 480 | 3 | 24 | 84.3 | 110 |
| 07217002 | HF3B5180CA1 | 80/60 | 273/204.7 | 240/208 | 3 | 24 | 192.8/166.7 | 250/225 |
| 07217102 | P3P5180CA1 | 80 | 273 | 480 | 3 | 24 | 96.3 | 125 |
| 07217602 | HF3B5190CA1 | 90/67.5 | 307.1/230.4 | 240/208 | 3 | 24 | 216.9/187.5 | 300/250 |
| 07217702 | P3P5190CA1 | 90 | 307.1 | 480 | 3 | 24 | 108.4 | 150 |
| 07218202 | HF3B51100CA1 | 100/75 | 341.3/256 | 240/208 | 3 | 24 | 241/208.3 | 350/300 |
| 07218302 | P3P51100CA1 | 100 | 341.3 | 480 | 3 | 24 | 120.4 | 175 |

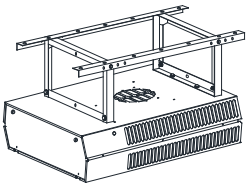
# *Recommended Control Options, Control Accessory Options, & Control Accessories*



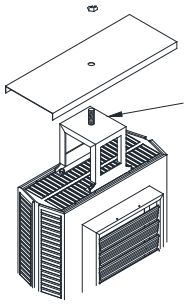
| A5100 BRACKET | | | |
|---|---|---|---|
| | HEATER KW | BRACKET | WT (LBS) |
| WALL BRACKET & HEATER BRACKET | 3.3 - 5 | A5105 | 9 |
| | 7.5 - 20 | A5120 | 13 |
| | 25 - 50 | A5150 | 16 |
| | 60 - 70 | A5175 | 25 |
| | 80 - 100 | A51100 | 36 |



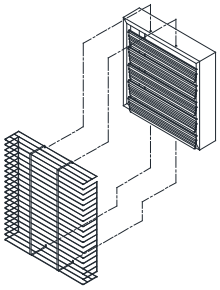
| B5100 BRACKET | | | |
|---|---|---|---|
| | HEATER KW | BRACKET | WT (LBS) |
| HEATER BRACKET | 3.3 -20 | B5105 | 2 |
| | 25 - 50 | B5150 | 3 |
| | 60 - 70 | B5170 | 5 |
| | 80 - 100 | B51100 | 12 |



| V5100 BRACKET | | | |
|---|---|---|---|
| | HEATER KW | BRACKET | WT (LBS) |
| VERTICAL MOUNT BRACKET | 3.3 - 5 | V5105 | 9 |
| | 7.5 - 20 | V5120 | 13 |
| | 25 - 50 | V5150 | 16 |
| | 60 - 70 | V5175 | 19 |
| | 80 - 100 | V51100 | 22 |



| DS5100 DUST SHIELD | | | |
|---|---|---|---|
| | HEATER KW | BRACKET | WT (LBS) |
| DUST SHIELD | 3.3 - 5 | DS5105 | 3 |
| | 7.5 - 20 | DS5120 | 4 |
| | 25 - 50 | DS5150 | 5 |
| | 60 - 70 | DS5175 | 10 |
| | 80 - 100 | DS51100 | 17 |



| OFG5100 FAN GUARD | | | |
|---|---|---|---|
| | HEATER KW | BRACKET | WT (LBS) |
| FAN GUARD | 3.3 - 5 | OFG5101 | 1 |
| | 7.5 - 20 | OFG5102 | 1 |
| | 25 - 50 | OFG5103 | 2 |
| | 60 - 70 | OFG5175 | 4 |
| | 80 - 100 | OFG51100 | 5 |

| DISCONNECT SWITCH | DISCONNECT SWITCH MUST BE DERATED FOR USE AT MAX 80% OF TOTAL HEATER RATED AMPERAGE |
|---|---|

| CONTROL ACCESSORY OPTIONS - FIELD INSTALLED IN HEATER | | |
|---|---|---|
| MFG CATALOG NUMBER | MFG MODEL NUMBER | DESCRIPTION |
| POWER DISCONNECT SWITCH* | | |
| 07171502 | DCS202 / 5100 | 20 AMP; 0-16 Amps |
| 07171602 | DCS403 / 5100 | 40 AMP; 16.1-32 Amps |
| 07171702 | DCS603 / 5100 | 60 AMP; 32.1-48 Amps |
| 07171802 | DCS1003 /5100 | 100 AMP; 48.1-80 Amps |
| *Disconnect must be rated at 80% of total ampload. | | |
| LINE VOLTAGE THERMOSTAT | | |
| 06927702 | T5100 | SPST; PILOT DUTY 25 AMP, 24-277V , (40-110 °F) |
| 06927802 | T5102 | DPST; LINE DUTY 25 AMP 120-277V 25 AMP, 120-277V , (40-120 °F) |
| LOW VOLTAGE THERMOSTAT (ALL CA1 MODELS) | | |
| 06927702 | T5100 | SPST; LOW VOLT/PILOT DUTY; 125VA; (3.3-20 KW UNITS) , (40-110 °F) |
| 06927902 | T5122 | 2-STAGE; LOW VOLT; 125VA; (25-50 KW UNITS) , (40-100 °F) |
| STRATIFICATION THERMOSTAT | | |
| 06928002 | TC5102 | SPST; LOW VOLT/PILOT DUTY (W/RELAY) ; 70-130° |
| 06928102 | TC5103 | SPST; LINE DUTY; AMP 120-240V; 70-130° |
| SUMMER FAN SWITCH | | |
| 06928202 | FS5101 | SPST; LINE VOLT ;120-277V |
| 06928302 | FS5102 | SPST; LINE VOLT ; 480-600V |

| CONTROL ACCESSORIES - REMOTE WALL MOUNTED | | |
|---|---|---|
| MFG CATALOG NUMBER | MFG MODEL NUMBER | DESCRIPTION |
| LINE VOLTAGE THERMOSTAT | | |
| 05029102 | S2025H10AA** | SP; 25 Amp; 120-277V; 50-90°F |
| 05238902 | ET9STS** | SP; 22 Amp; 120-277V; 50-90°F |
| 06910902 | TW1512* | DPST; 25 AMP; 120-277V; 50-90°F |
| LOW VOLTAGE THERMOSTATS (ALL CA1 MODELS) | | |
| 05260102 | RK120EAA | 24VAC or MV; 50°-90°F (3.3-20kW) |
| 07425202 | PRO1 T721 | DIGITAL; 24V; 2 STAGE; 44°-90°F |
| STRATIFICATION THERMOSTAT | | |
| 06911002 | TC1602 | SPST; LINE VOLT; 120-277V; 70-140° |
| SUMMER FAN SWITCH | | |
| 06926202 | FSW5111 | SPST; LINE VOLT; 120-277V |
| 06926302 | FSW5112 | SPST; LOW VOLT; (W/RELAY) |
| LOW VOLTAGE THERMOSTAT & SUMMER FAN SWITCH (W/RELAY) (CA1 MODELS) | | |
| 06926402 | TFS5101 | SPST; LOW VOLT; 50-90° |
| 06926502 | TFS5102 | ELECTRONIC; 1 OR 2 STAGE; LOW VOLT; 40° - 90°F |

*Use DPST double line break thermostat for 3-phase line voltage control in non "C" models. **Any SPST thermostat of sufficient amperage may be substituted.
*** For complete specifications, refer to page 108.

## *Recommended Control Options, Control Accessory Options, & Control Accessories*

| MFG MODEL NUMBER | DISCONNECT SWITCH | | THERMOSTAT | | SUMMER FAN SWITCH | | THERMOSTAT & SUMMER FAN SWITCH WALL MOUNTED | STRATIFICATION THERMOSTAT | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 Ø | 3 Ø | IN-BUILT | WALL MOUNTED | IN-BUILT | WALL MOUNTED | | IN-BUILT | WALL MOUNTED |
| F1F5103N | DCS 202 | NA | T5100 | ET9STS | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| HF1B5103N | DCS 202 | NA | T5100 | ET9STS | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| F2F5103N | DCS 204 | NA | T5100 | ET9STS | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| F2F5103N | NA | DCS 403 | T5102 | TW 1512 | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| HF2B5103N | DCS 202 | NA | T5100 | ET9STS | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| HF2B5103N | NA | DCS 403 | T5102 | TW 1512 | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| HF2B5103N | DCS 202 | NA | T5100 | ET9STS | FS5101 | FSW5111 | NA | NA | TC1602 |
| P3P5103CA1N | NA | DCS 403 | T5100 | RK120EAA | FS5102 | FSW5112 | TFS5101 | TC5102 | NA |
| F1F5105N | DCS 403 | NA | T5100 | S2025 | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| HF1B5105N | DCS 403 | NA | T5100 | ET9STS | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| F2F5105N | DCS 403 | NA | T5100 | S2025 | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| F2F5105N | NA | DCS 403 | T5102 | TW 1512 | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| HF2B5105N | DCS 403 | NA | T5100 | ET9STS | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| HF2B5105N | NA | DCS 403 | T5102 | TW 1512 | FS5101 | FSW5111 | NA | TC5103 | TC1602 |
| G1G5105N | DCS 403 | NA | T5100 | S2025 | FS5101 | FSW5111 | NA | NA | TC1602 |
| P3P5105CA1N | NA | DCS 403 | T5100 | RK120EAA | FS5102 | FSW5112 | TFS5101 | TC5102 | NA |
| F2F5107CA1L | DCS 603 | NA | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| F2F5107CA1L | NA | DCS 603 | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF2B 5107CA1L | DCS 403 | NA | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF2B 5107CA1L | NA | DCS 403 | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| G1G5107CA1L | DCS 403 | NA | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | NA | TC1602 |
| P3P5107CA1L | NA | DCS 403 | T5100 | RK120EAA | FS5102 | FSW5112 | TFS5101 | TC5102 | NA |
| F2F5110CA1L | DCS 603 | NA | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| F2F5110CA1L | NA | DCS 603 | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF2B5110CA1L | DCS 603 | NA | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF2B5110CA1L | NA | DCS 603 | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| G1G5110CA1N | DCS 603 | NA | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | NA | TC1602 |
| P3P5110CA1N | NA | DCS 403 | T5100 | RK120EAA | FS5102 | FSW5112 | TFS5101 | TC5103 | NA |
| F3F5115CA1L | NA | DCS 603 | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF3B5115CA1L | NA | DCS 603 | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| P3P5115CA1L | NA | DCS 403 | T5100 | RK120EAA | FS5102 | FSW5112 | TFS5101 | TC5103 | NA |
| HF3B5120CA1L | NA | DCS 603 | T5100 | RK120EAA | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| P3P5120CA1N | NA | DCS 403 | T5100 | RK120EAA | FS5102 | FSW5112 | TFS5101 | TC5103 | NA |
| F3F5125CA1L | NA | DCS 1003 | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF3B5125CA1L | NA | DCS 1003 | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| P3P5125CA1N | NA | DCS 403 | T5100 | T721 | FS5102 | FSW5112 | TFS5101 | TC5103 | NA |
| F3F5130CA1L | NA | NA | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF3B5130CA1L | NA | DCS 1003 | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| P3P5130CA1N | NA | DCS 603 | T5100 | T721 | FS5102 | FSW5112 | TFS5101 | TC5103 | NA |
| F3F5140CA1L | NA | NA | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF3B5140CA1L | NA | NA | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| P3P5140CA1N | NA | DCS 603 | T5100 | T721 | FS5102 | FSW5112 | TFS5101 | TC5103 | NA |
| F3F5150CA1L | NA | NA | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| HF3B5150CA1L | NA | NA | T5100 | T721 | FS5101 | FSW5111 | TFS5101 | TC5103 | TC1602 |
| P3P5150CA1N | NA | DCS 1003 | T5100 | T721 | FS5102 | FSW5112 | TFS5101 | TC5103 | NA |

## *Recommended Control Options 50kW - 100 kW Models*

| MFG CATALOG NUMBER | MFG MODEL NUMBER | STEP CONTROLLER | REMOTE THERMOSTAT | SUMMER FAN SWITCH | | STRATIFICATION THERMOSTAT | |
|---|---|---|---|---|---|---|---|
| | | | | WALL | BUILT-IN | WALL | BUILT-IN |
| | | CATALOG/ MODEL | CATALOG/ MODEL | CATALOG/ MODEL | MODEL | CATALOG/ MODEL | CATALOG/ MODEL |
| 07213402 | F3F5160CA1 | 05232302 SC5100 | 44046001 VT7225 (0-10VDC) | 06926302 FSW5112 | FS5101 | 06926402 TFS5101 | 06928002 TC5102 |
| 07213502 | HF3B5160CA1 | | | | FS5101 | | |
| 07213602 | P3P5160CA1 | | | | FS5102 | | |
| 07214202 | F3F5170CA1 | | | | FS5101 | | |
| 07214302 | HF3B5170CA1 | | | | FS5101 | | |
| 07214402 | P3P5170CA1 | | | | FS5102 | | |
| 07217002 | HF3B5180CA1 | | | | FS5101 | | |
| 07217102 | P3P5180CA1 | | | | FS5102 | | |
| 07217602 | HF3B5190CA1 | | | | FS5101 | | |
| 07217702 | P3P5190CA1 | | | | FS5102 | | |
| 07218202 | HF3B51100CA1 | | | | FS5101 | | |
| 07218302 | P3P51100CA1 | | | | FS5102 | | |